

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (MS CS, 2.5+ years) building enterprise-grade AI agents for complex document workflows from inception to production. Designed a multi-agent insurance claims processing system with schema-validated agent contracts, RAG-grounded regulatory validation, and explainable audit-ready risk scoring - directly applicable to FurtherAI's insurance AI Teammates automating document processing, data entry, and workflow navigation. Delivered >30% hallucination reduction and 25% agent stability improvement in production healthcare environments. Eligible for visa sponsorship (F-1 OPT).

CORE COMPETENCIES

Enterprise AI Agent Development • Multi-Agent Pipeline Architecture • Document Processing AI • RAG & Regulatory Validation • Schema-Validated Agent Contracts • LangChain / LlamaIndex • Python / FastAPI / Docker • Customer Workflow Translation • Production Deployment & Hardening

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **enterprise AI agents** for multi-step healthcare workflows - eligibility checks, care workflow navigation, documentation clarification - with structured reasoning and tool discipline; improved stability ~**25%**, reduced hallucinations >**30%**.
- Developed **RAG systems** for clinical document retrieval and structured data extraction from EHR records; improved precision ~**35%**, grounding alignment exceeded **90%**.
- Built **predictive ML pipelines** (scikit-learn, XGBoost) for patient engagement and risk stratification; improved recall **15-20%** on high-risk categories.
- Packaged AI as **FastAPI/Flask RESTful services** with Docker; data entry automation and system integration for healthcare enterprise; reduced post-deploy defects ~**30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, evaluation audit trails.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Document AI

- Built transformer-based hierarchical summarization for 5,000+ sessions; **60-70% review time reduction**, ~**85%** highlight alignment.
- RAG document Q&A: reduced hallucinations ~**30%**, raised accuracy >**85%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Enterprise ERP SQL optimization, data entry automation, Jenkins CI/CD, Grafana dashboards.

KEY PROJECTS

Agentic Insurance Claims Processing & Fraud Risk Intelligence System

Built a **multi-agent pipeline** for insurance claims automation: Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring. **Schema-validated JSON contracts between agents** prevent cascading hallucinations. **RAG-grounded CPT/ICD validation** against vector-indexed regulatory documents. ANN-based duplicate detection for claim deduplication. Explainable risk scoring with **audit-ready reasoning traces** - identical architecture to FurtherAI's insurance workflow AI Teammates.

Agentic Pixel Character Synthesis - LLM Orchestration

LLM agent orchestrator decomposes high-level prompts into multi-step Stable Diffusion + LoRA + ControlNet generation pipeline. Demonstrates agentic decomposition of complex tasks into tool-calling workflows - core FDE skill.

Manga Lens - Multi-Provider AI Production Extension

Production Chrome extension with four AI providers (Claude, GPT-4o, Ollama). End-to-end engineering: content scripts, service workers, provider abstraction, 7-day cache, 29 sites. Shows full-stack ownership from customer workflow → production system.

Patient Records Management - Healthcare CRUD Application

Bottle + SQLite healthcare app with normalized relational schema, cascade deletes, and aggregate patient/diagnosis queries. Enterprise data entry automation for medical records - direct FurtherAI insurance data entry parallel.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, React, SQL, JavaScript, PHP

LLM & AI: RAG, agentic workflows, multi-agent orchestration, LangChain, LlamaIndex, prompt engineering, vector search, structured outputs, guardrails

ML: PyTorch, scikit-learn, XGBoost, HuggingFace Transformers

Infra: Docker, FastAPI, Flask, Jenkins, REST APIs, Grafana

Work Authorization: F-1 OPT (US-authorized now); visa sponsorship available per FurtherAI